Curriculum Coverage Annotation — Human Evaluation Sheet (0–10)

Instructions: each item pairs a skill demanded in Cambodian IT job postings with one learning outcome from an ITC course (Years 3–5, Software Engineering programme). Judge from the text alone: **how well does this course outcome teach this skill?** The course each outcome belongs to is shown in the last column for context. A glossary is on the last page.

Score 0–10: 0 = not taught here · 5 = related / foundation, but not this skill itself · 10 = fully teaches this exact skill.

Rows 1–30 are the core block — please complete at least these 30. After that, continue as far as you have time; every additional rating helps.

250 pairs · annotator: _____ · date: _____ · role (circle): lecturer / senior student / classmate / industry

#	Market Skill	Course Learning Outcome	Score (0–10) — circle one	Note	Course (Year)
1	**security monitoring**	Business Model Canvas	0 1 2 3 4 5 6 7 8 9 10		Soft Skills (Y3)
2	**firewall configuration**	Integrity Constraints	0 1 2 3 4 5 6 7 8 9 10		Information System Analysis And Design (Y3)
3	**postgresql database**	Link Aggregation (LAG)	0 1 2 3 4 5 6 7 8 9 10		Networks Ii (Y4)
4	**data analysis**	Mobile Operating Systems Fundamentals	0 1 2 3 4 5 6 7 8 9 10		Introduction To Mobile Application Development (Y4)
5	**data analysis**	Data Models	0 1 2 3 4 5 6 7 8 9 10		Information System Analysis And Design (Y3)
6	**it infrastructure management**	Security Technology Deployment	0 1 2 3 4 5 6 7 8 9 10		Network Security (Y5)
7	**software development**	Hybrid App Development	0 1 2 3 4 5 6 7 8 9 10		Introduction To Mobile Application Development (Y4)
8	**docker containerization**	Database Partitioning	0 1 2 3 4 5 6 7 8 9 10		Advanced Database And Database Management Systems (Y4)
9	**postgresql database**	Database Queries	0 1 2 3 4 5 6 7 8 9 10		Data Mining (Y5)
10	**system integration**	System Reliability Engineering	0 1 2 3 4 5 6 7 8 9 10		Network Design (Y4)
11	**data analysis**	Boolean Algebra	0 1 2 3 4 5 6 7 8 9 10		Combinational And Sequential Logics (Y3)
12	**network administration**	Business Model Canvas	0 1 2 3 4 5 6 7 8 9 10		Soft Skills (Y3)
13	**technical support**	Network Maintenance and Troubleshooting	0 1 2 3 4 5 6 7 8 9 10		Network Design (Y4)
14	**performance tuning**	Resource Estimation	0 1 2 3 4 5 6 7 8 9 10		It Project Management (Y5)
15	**restful api development**	Hybrid App Development	0 1 2 3 4 5 6 7 8 9 10		Introduction To Mobile Application Development (Y4)
16	**ci/cd pipelines**	Machine Learning Pipeline	0 1 2 3 4 5 6 7 8 9 10		Introduction To Artificial Intelligence (Y5)
17	**security monitoring**	Wi-Fi Security Protocols	0 1 2 3 4 5 6 7 8 9 10		Networks Ii (Y4)
18	**mysql database**	Database Management Systems (DBMS)	0 1 2 3 4 5 6 7 8 9 10		Information System Analysis And Design (Y3)
19	**aws cloud platforms**	Database Management Systems (DBMS)	0 1 2 3 4 5 6 7 8 9 10		Information System Analysis And Design (Y3)
20	**git version control**	Model Evaluation	0 1 2 3 4 5 6 7 8 9 10		Data Mining (Y5)
21	**software development**	Software as a Service (SaaS)	0 1 2 3 4 5 6 7 8 9 10		Cloud Computing (Y5)
22	**incident response**	Counters	0 1 2 3 4 5 6 7 8 9 10		Combinational And Sequential Logics (Y3)
23	**firewall configuration**	Web Server Configuration	0 1 2 3 4 5 6 7 8 9 10		Internet Programming Ii (Y4)
24	**technical documentation**	Security Standards and Policies	0 1 2 3 4 5 6 7 8 9 10		Information Security (Y5)
25	**incident response**	Exception Handling	0 1 2 3 4 5 6 7 8 9 10		Object Oriented Programming (Y3)
26	**security compliance**	Cloud Computing Security	0 1 2 3 4 5 6 7 8 9 10		Cloud Computing (Y5)
27	**database management**	Pattern Recognition and Classification	0 1 2 3 4 5 6 7 8 9 10		Image Processing (Y5)
28	**system maintenance**	Security Solutions for System Support and Operations	0 1 2 3 4 5 6 7 8 9 10		Information Security (Y5)
29	**server management**	Project Issue Management	0 1 2 3 4 5 6 7 8 9 10		It Project Management (Y5)
30	**git version control**	Business Model Canvas	0 1 2 3 4 5 6 7 8 9 10		Soft Skills (Y3)

#	Market Skill	Course Learning Outcome	Score (0–10) — circle one	Note	Course (Year)
31	**technical documentation**	Project Status Reporting	0 1 2 3 4 5 6 7 8 9 10		It Project Management (Y5)
32	**data analysis**	Data Analysis	0 1 2 3 4 5 6 7 8 9 10		Research Methodology (Y3)
33	**network troubleshooting**	Network Devices	0 1 2 3 4 5 6 7 8 9 10		Network Security (Y5)
34	**git version control**	Transaction Control (Commit, Rollback, Locking)	0 1 2 3 4 5 6 7 8 9 10		Advanced Database And Database Management Systems (Y4)
35	**security monitoring**	Network Monitoring	0 1 2 3 4 5 6 7 8 9 10		Network Design (Y4)
36	**mysql database**	Database Indexing	0 1 2 3 4 5 6 7 8 9 10		Advanced Database And Database Management Systems (Y4)
37	**technical documentation**	Network Protocols and Standards	0 1 2 3 4 5 6 7 8 9 10		Network Security (Y5)
38	**security monitoring**	Database Security Policy Design	0 1 2 3 4 5 6 7 8 9 10		Advanced Database And Database Management Systems (Y4)
39	**performance tuning**	Cloud Workload Optimization	0 1 2 3 4 5 6 7 8 9 10		Cloud Computing (Y5)
40	**server management**	Network Management	0 1 2 3 4 5 6 7 8 9 10		Network Design (Y4)
41	**system integration**	Network Architectures	0 1 2 3 4 5 6 7 8 9 10		Introduction To Computer Systems And Networks (Y3)
42	**aws cloud platforms**	AWS	0 1 2 3 4 5 6 7 8 9 10		Cloud Computing (Y5)
43	**it infrastructure management**	Software Application Maintenance	0 1 2 3 4 5 6 7 8 9 10		Object Oriented Programming (Y3)
44	**network troubleshooting**	Project Issue Management	0 1 2 3 4 5 6 7 8 9 10		It Project Management (Y5)
45	**java programming**	Python	0 1 2 3 4 5 6 7 8 9 10		Image Processing (Y5)
46	**database management**	User Account Management	0 1 2 3 4 5 6 7 8 9 10		Systems And Networks Administration (Y4)
47	**security compliance**	Security Technology Deployment	0 1 2 3 4 5 6 7 8 9 10		Network Security (Y5)
48	**it infrastructure management**	Infrastructure Provisioning and Management with IaC	0 1 2 3 4 5 6 7 8 9 10		Software Development And It Operations Devops (Y4)
49	**restful api development**	Application Programming Interfaces (APIs)	0 1 2 3 4 5 6 7 8 9 10		Internet Programming Ii (Y4)
50	**database management**	Database Transaction Management	0 1 2 3 4 5 6 7 8 9 10		Advanced Database And Database Management Systems (Y4)
51	**it infrastructure management**	Network Management	0 1 2 3 4 5 6 7 8 9 10		Network Design (Y4)
52	**system integration**	Virtualization	0 1 2 3 4 5 6 7 8 9 10		Cloud Computing (Y5)
53	**data analysis**	Data Warehousing	0 1 2 3 4 5 6 7 8 9 10		Advanced Database And Database Management Systems (Y4)
54	**security compliance**	Mobile App Security	0 1 2 3 4 5 6 7 8 9 10		Introduction To Mobile Application Development (Y4)
55	**network troubleshooting**	Pitching	0 1 2 3 4 5 6 7 8 9 10		Soft Skills (Y3)
56	**performance tuning**	Task-Relevant Data Selection	0 1 2 3 4 5 6 7 8 9 10		Data Mining (Y5)
57	**java programming**	Application Programming Interfaces (APIs)	0 1 2 3 4 5 6 7 8 9 10		Internet Programming Ii (Y4)
58	**disaster recovery planning**	Disaster Recovery Planning	0 1 2 3 4 5 6 7 8 9 10		Network Security (Y5)
59	**postgresql database**	Database Partitioning	0 1 2 3 4 5 6 7 8 9 10		Advanced Database And Database Management Systems (Y4)
60	**security monitoring**	Project Progress Monitoring	0 1 2 3 4 5 6 7 8 9 10		It Project Management (Y5)
61	**software development**	Hypothesis Development	0 1 2 3 4 5 6 7 8 9 10		Research Methodology (Y3)
62	**ci/cd pipelines**	Distributed System Processes	0 1 2 3 4 5 6 7 8 9 10		Distributed Systems (Y4)
63	**security monitoring**	Network Monitoring Tools	0 1 2 3 4 5 6 7 8 9 10		Systems And Networks Administration (Y4)
64	**project management**	Network Management	0 1 2 3 4 5 6 7 8 9 10		Network Design (Y4)
65	**firewall configuration**	Firewall Technology	0 1 2 3 4 5 6 7 8 9 10		Network Security (Y5)
66	**network administration**	Access Control	0 1 2 3 4 5 6 7 8 9 10		Information Security (Y5)

#	Market Skill	Course Learning Outcome	Score (0–10) — circle one	Note	Course (Year)
67	**aws cloud platforms**	Apache Hadoop	0 1 2 3 4 5 6 7 8 9 10		Data Mining (Y5)
68	**system integration**	Distributed System Processes	0 1 2 3 4 5 6 7 8 9 10		Distributed Systems (Y4)
69	**microsoft office proficiency**	Business Model Canvas	0 1 2 3 4 5 6 7 8 9 10		Soft Skills (Y3)
70	**api integration**	Hybrid App Architecture	0 1 2 3 4 5 6 7 8 9 10		Introduction To Mobile Application Development (Y4)
71	**firewall configuration**	Business Model Canvas	0 1 2 3 4 5 6 7 8 9 10		Soft Skills (Y3)
72	**code reviews**	Infrastructure as Code (IaC)	0 1 2 3 4 5 6 7 8 9 10		Software Development And It Operations Devops (Y4)
73	**security monitoring**	Sensor Data Collection	0 1 2 3 4 5 6 7 8 9 10		Data Mining (Y5)
74	**technical documentation**	Monitoring and Logging Strategies	0 1 2 3 4 5 6 7 8 9 10		Software Development And It Operations Devops (Y4)
75	**server management**	Load Balancing	0 1 2 3 4 5 6 7 8 9 10		Software Development And It Operations Devops (Y4)
76	**java programming**	C++ Programming	0 1 2 3 4 5 6 7 8 9 10		Algorithm And Programming (Y3)
77	**software development**	C++ Programming	0 1 2 3 4 5 6 7 8 9 10		Algorithm And Programming (Y3)
78	**performance tuning**	Client-Server Architecture	0 1 2 3 4 5 6 7 8 9 10		Internet Programming Ii (Y4)
79	**security compliance**	Network Security Design	0 1 2 3 4 5 6 7 8 9 10		Network Design (Y4)
80	**api integration**	Distributed Application Design and Implementation	0 1 2 3 4 5 6 7 8 9 10		Distributed Systems (Y4)
81	**network troubleshooting**	Network Infrastructure Design	0 1 2 3 4 5 6 7 8 9 10		Networks Ii (Y4)
82	**system maintenance**	Network Troubleshooting	0 1 2 3 4 5 6 7 8 9 10		Introduction To Computer Systems And Networks (Y3)
83	**code reviews**	Algorithm Analysis	0 1 2 3 4 5 6 7 8 9 10		Discrete Mathematics (Y3)
84	**ci/cd pipelines**	DevOps Methodology	0 1 2 3 4 5 6 7 8 9 10		Cloud Computing (Y5)
85	**api integration**	Interfaces	0 1 2 3 4 5 6 7 8 9 10		Object Oriented Programming (Y3)
86	**performance tuning**	Network Performance Monitoring	0 1 2 3 4 5 6 7 8 9 10		Systems And Networks Administration (Y4)
87	**project management**	Project Team Management	0 1 2 3 4 5 6 7 8 9 10		It Project Management (Y5)
88	**network troubleshooting**	Network Maintenance and Troubleshooting	0 1 2 3 4 5 6 7 8 9 10		Network Design (Y4)
89	**server management**	Database Management Systems (DBMS)	0 1 2 3 4 5 6 7 8 9 10		Data Mining (Y5)
90	**technical documentation**	Text Processing	0 1 2 3 4 5 6 7 8 9 10		Natural Language Processing (Y5)
91	**software development**	Agile Project Management	0 1 2 3 4 5 6 7 8 9 10		It Project Management (Y5)
92	**network administration**	Client-Server Architecture	0 1 2 3 4 5 6 7 8 9 10		Internet Programming Ii (Y4)
93	**data analysis**	Text Feature Extraction	0 1 2 3 4 5 6 7 8 9 10		Natural Language Processing (Y5)
94	**software development**	Software Requirements Analysis	0 1 2 3 4 5 6 7 8 9 10		Object Oriented Programming (Y3)
95	**technical documentation**	Data Integrity and Data Quality Documentation	0 1 2 3 4 5 6 7 8 9 10		Information System Analysis And Design (Y3)
96	**security monitoring**	Cryptography	0 1 2 3 4 5 6 7 8 9 10		Discrete Mathematics (Y3)
97	**api integration**	Continuous Integration Tools Setup and Configuration	0 1 2 3 4 5 6 7 8 9 10		Software Development And It Operations Devops (Y4)
98	**security compliance**	Network Security Measures	0 1 2 3 4 5 6 7 8 9 10		Systems And Networks Administration (Y4)
99	**security monitoring**	Mobile App Security	0 1 2 3 4 5 6 7 8 9 10		Introduction To Mobile Application Development (Y4)
100	**network administration**	Network Administrator Responsibilities	0 1 2 3 4 5 6 7 8 9 10		Systems And Networks Administration (Y4)
101	**system integration**	Project Issue Management	0 1 2 3 4 5 6 7 8 9 10		It Project Management (Y5)
102	**database management**	Relations and Relational Database Modeling	0 1 2 3 4 5 6 7 8 9 10		Discrete Mathematics (Y3)

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103	**database management**	Database Management Systems (DBMS)	0 1 2 3 4 5 6 7 8 9 10		Data Mining (Y5)
104	**vmware virtualization**	Distributed System Processes	0 1 2 3 4 5 6 7 8 9 10		Distributed Systems (Y4)
105	**network administration**	Infrastructure as a Service (IaaS)	0 1 2 3 4 5 6 7 8 9 10		Cloud Computing (Y5)
106	**network troubleshooting**	Network Troubleshooting	0 1 2 3 4 5 6 7 8 9 10		Systems And Networks Administration (Y4)
107	**it infrastructure management**	Distributed Systems Architectures	0 1 2 3 4 5 6 7 8 9 10		Distributed Systems (Y4)
108	**performance tuning**	Distributed System Performance Analysis	0 1 2 3 4 5 6 7 8 9 10		Distributed Systems (Y4)
109	**technical documentation**	Data Collection Methods	0 1 2 3 4 5 6 7 8 9 10		Research Methodology (Y3)
110	**system integration**	Hybrid App Architecture	0 1 2 3 4 5 6 7 8 9 10		Introduction To Mobile Application Development (Y4)
111	**security monitoring**	Laravel Security Features	0 1 2 3 4 5 6 7 8 9 10		Internet Programming Ii (Y4)
112	**code reviews**	C++ Programming	0 1 2 3 4 5 6 7 8 9 10		Algorithm And Programming (Y3)
113	**it infrastructure management**	Pointwise Mutual Information (PMI)	0 1 2 3 4 5 6 7 8 9 10		Natural Language Processing (Y5)
114	**mysql database**	Database Seeding	0 1 2 3 4 5 6 7 8 9 10		Internet Programming Ii (Y4)
115	**it infrastructure management**	Project Issue Management	0 1 2 3 4 5 6 7 8 9 10		It Project Management (Y5)
116	**database management**	State Tables	0 1 2 3 4 5 6 7 8 9 10		Combinational And Sequential Logics (Y3)
117	**server management**	Distributed System Processes	0 1 2 3 4 5 6 7 8 9 10		Distributed Systems (Y4)
118	**security monitoring**	Information Security Threat Trend Monitoring	0 1 2 3 4 5 6 7 8 9 10		Information Security (Y5)
119	**system integration**	Distributed Database Management	0 1 2 3 4 5 6 7 8 9 10		Advanced Database And Database Management Systems (Y4)
120	**firewall configuration**	Firewall Rules	0 1 2 3 4 5 6 7 8 9 10		Systems And Networks Administration (Y4)
121	**system maintenance**	Virtualization	0 1 2 3 4 5 6 7 8 9 10		Cloud Computing (Y5)
122	**security compliance**	Security Models	0 1 2 3 4 5 6 7 8 9 10		Distributed Systems (Y4)
123	**system maintenance**	Information System Components	0 1 2 3 4 5 6 7 8 9 10		Information System Analysis And Design (Y3)
124	**system integration**	Continuous Integration Tools Setup and Configuration	0 1 2 3 4 5 6 7 8 9 10		Software Development And It Operations Devops (Y4)
125	**it infrastructure management**	Distributed Database Management	0 1 2 3 4 5 6 7 8 9 10		Advanced Database And Database Management Systems (Y4)
126	**vmware virtualization**	Scalable Software Design	0 1 2 3 4 5 6 7 8 9 10		Object Oriented Programming (Y3)
127	**postgresql database**	C++ Programming	0 1 2 3 4 5 6 7 8 9 10		Algorithm And Programming (Y3)
128	**system integration**	Security Technology Deployment	0 1 2 3 4 5 6 7 8 9 10		Network Security (Y5)
129	**network administration**	Infrastructure Provisioning and Management with IaC	0 1 2 3 4 5 6 7 8 9 10		Software Development And It Operations Devops (Y4)
130	**java programming**	Algorithm Analysis	0 1 2 3 4 5 6 7 8 9 10		Discrete Mathematics (Y3)
131	**project management**	Risk Management	0 1 2 3 4 5 6 7 8 9 10		Information Security (Y5)
132	**database management**	Database Management Systems (DBMS)	0 1 2 3 4 5 6 7 8 9 10		Information System Analysis And Design (Y3)
133	**server management**	Software Application Maintenance	0 1 2 3 4 5 6 7 8 9 10		Object Oriented Programming (Y3)
134	**technical documentation**	Truth-Table Derivation From Logic Diagrams	0 1 2 3 4 5 6 7 8 9 10		Electronics (Y3)
135	**database management**	Network Management	0 1 2 3 4 5 6 7 8 9 10		Network Design (Y4)
136	**data analysis**	Correlation Analysis	0 1 2 3 4 5 6 7 8 9 10		Data Mining (Y5)
137	**vmware virtualization**	Infrastructure Provisioning and Management with IaC	0 1 2 3 4 5 6 7 8 9 10		Software Development And It Operations Devops (Y4)
138	**system integration**	Security Solutions for System Support and Operations	0 1 2 3 4 5 6 7 8 9 10		Information Security (Y5)

#	Market Skill	Course Learning Outcome	Score (0–10) — circle one	Note	Course (Year)
139	**data analysis**	Principal Component Analysis (PCA)	0 1 2 3 4 5 6 7 8 9 10		Introduction To Artificial Intelligence (Y5)
140	**technical support**	Network Troubleshooting	0 1 2 3 4 5 6 7 8 9 10		Systems And Networks Administration (Y4)
141	**vmware virtualization**	Cloud Computing Security	0 1 2 3 4 5 6 7 8 9 10		Network Security (Y5)
142	**mysql database**	Resource Estimation	0 1 2 3 4 5 6 7 8 9 10		It Project Management (Y5)
143	**security monitoring**	Data Anomaly Resolution	0 1 2 3 4 5 6 7 8 9 10		Information System Analysis And Design (Y3)
144	**software development**	Distributed System Processes	0 1 2 3 4 5 6 7 8 9 10		Distributed Systems (Y4)
145	**java programming**	Database Queries	0 1 2 3 4 5 6 7 8 9 10		Data Mining (Y5)
146	**project management**	Risk Management and Assessment	0 1 2 3 4 5 6 7 8 9 10		Network Security (Y5)
147	**system maintenance**	Network Troubleshooting	0 1 2 3 4 5 6 7 8 9 10		Systems And Networks Administration (Y4)
148	**server management**	Access Control and Permissions	0 1 2 3 4 5 6 7 8 9 10		Network Security (Y5)
149	**java programming**	Java	0 1 2 3 4 5 6 7 8 9 10		Object Oriented Programming (Y3)
150	**database management**	Data Preprocessing	0 1 2 3 4 5 6 7 8 9 10		Introduction To Artificial Intelligence (Y5)
151	**firewall configuration**	Firewall Configuration	0 1 2 3 4 5 6 7 8 9 10		Introduction To Computer Systems And Networks (Y3)
152	**network administration**	Network Management	0 1 2 3 4 5 6 7 8 9 10		Network Design (Y4)
153	**it infrastructure management**	Production Deployment	0 1 2 3 4 5 6 7 8 9 10		Internet Programming Ii (Y4)
154	**network troubleshooting**	Networking Principles for Distributed Systems	0 1 2 3 4 5 6 7 8 9 10		Distributed Systems (Y4)
155	**it infrastructure management**	Information System Components	0 1 2 3 4 5 6 7 8 9 10		Information System Analysis And Design (Y3)
156	**data analysis**	Algorithm Analysis	0 1 2 3 4 5 6 7 8 9 10		Discrete Mathematics (Y3)
157	**it infrastructure management**	Network Performance Monitoring	0 1 2 3 4 5 6 7 8 9 10		Systems And Networks Administration (Y4)
158	**network troubleshooting**	Infrastructure Monitoring	0 1 2 3 4 5 6 7 8 9 10		Software Development And It Operations Devops (Y4)
159	**incident response**	Alerting	0 1 2 3 4 5 6 7 8 9 10		Software Development And It Operations Devops (Y4)
160	**system integration**	Database Management Systems (DBMS)	0 1 2 3 4 5 6 7 8 9 10		Data Mining (Y5)
161	**security compliance**	Security Compliance	0 1 2 3 4 5 6 7 8 9 10		Information Security (Y5)
162	**data analysis**	Truth Table Analysis	0 1 2 3 4 5 6 7 8 9 10		Electronics (Y3)
163	**system maintenance**	Project Issue Management	0 1 2 3 4 5 6 7 8 9 10		It Project Management (Y5)
164	**project management**	Group Policy Management	0 1 2 3 4 5 6 7 8 9 10		Systems And Networks Administration (Y4)
165	**system integration**	Virtualization With VMware Workstation	0 1 2 3 4 5 6 7 8 9 10		Systems And Networks Administration (Y4)
166	**code reviews**	Binary Code Encoding and Decoding	0 1 2 3 4 5 6 7 8 9 10		Electronics (Y3)
167	**vmware virtualization**	Virtual Network Design	0 1 2 3 4 5 6 7 8 9 10		Introduction To Computer Systems And Networks (Y3)
168	**server management**	Web Server Configuration	0 1 2 3 4 5 6 7 8 9 10		Internet Programming Ii (Y4)
169	**technical documentation**	Feature Encoding	0 1 2 3 4 5 6 7 8 9 10		Introduction To Artificial Intelligence (Y5)
170	**data analysis**	Pattern Recognition and Classification	0 1 2 3 4 5 6 7 8 9 10		Image Processing (Y5)
171	**server management**	Ubuntu Server	0 1 2 3 4 5 6 7 8 9 10		Systems And Networks Administration (Y4)
172	**incident response**	Infrastructure as a Service (IaaS)	0 1 2 3 4 5 6 7 8 9 10		Cloud Computing (Y5)
173	**technical documentation**	State Diagrams	0 1 2 3 4 5 6 7 8 9 10		Combinational And Sequential Logics (Y3)
174	**network administration**	Networking Principles for Distributed Systems	0 1 2 3 4 5 6 7 8 9 10		Distributed Systems (Y4)
175	**security monitoring**	Packet Tracer	0 1 2 3 4 5 6 7 8 9 10		Introduction To Computer Systems And Networks (Y3)

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176	**incident response**	Disaster Recovery Planning	0 1 2 3 4 5 6 7 8 9 10		Network Security (Y5)
177	**system integration**	Information System Components	0 1 2 3 4 5 6 7 8 9 10		Information System Analysis And Design (Y3)
178	**server management**	Database Management Systems (DBMS)	0 1 2 3 4 5 6 7 8 9 10		Information System Analysis And Design (Y3)
179	**security compliance**	Wi-Fi Security Protocols	0 1 2 3 4 5 6 7 8 9 10		Networks Ii (Y4)
180	**performance tuning**	Performance Evaluation and Overfitting/Underfitting Diagnosis	0 1 2 3 4 5 6 7 8 9 10		Introduction To Artificial Intelligence (Y5)
181	**security compliance**	Integrity Constraints	0 1 2 3 4 5 6 7 8 9 10		Information System Analysis And Design (Y3)
182	**docker containerization**	Docker	0 1 2 3 4 5 6 7 8 9 10		Software Development And It Operations Devops (Y4)
183	**security monitoring**	Infrastructure Monitoring	0 1 2 3 4 5 6 7 8 9 10		Software Development And It Operations Devops (Y4)
184	**vmware virtualization**	Virtualization	0 1 2 3 4 5 6 7 8 9 10		Cloud Computing (Y5)
185	**performance tuning**	Network Optimization	0 1 2 3 4 5 6 7 8 9 10		Network Design (Y4)
186	**server management**	Network Device Management	0 1 2 3 4 5 6 7 8 9 10		Introduction To Computer Systems And Networks (Y3)
187	**performance tuning**	Asynchronous Counters	0 1 2 3 4 5 6 7 8 9 10		Combinational And Sequential Logics (Y3)
188	**system maintenance**	Software Application Maintenance	0 1 2 3 4 5 6 7 8 9 10		Object Oriented Programming (Y3)
189	**performance tuning**	Concurrency Control	0 1 2 3 4 5 6 7 8 9 10		Advanced Database And Database Management Systems (Y4)
190	**code reviews**	Literature Review	0 1 2 3 4 5 6 7 8 9 10		Research Methodology (Y3)
191	**java programming**	String Comparison	0 1 2 3 4 5 6 7 8 9 10		Natural Language Processing (Y5)
192	**aws cloud platforms**	Distributed Systems Architectures	0 1 2 3 4 5 6 7 8 9 10		Distributed Systems (Y4)
193	**performance tuning**	Application Performance Monitoring (APM)	0 1 2 3 4 5 6 7 8 9 10		Software Development And It Operations Devops (Y4)
194	**git version control**	Git	0 1 2 3 4 5 6 7 8 9 10		Software Development And It Operations Devops (Y4)
195	**network administration**	Network Device Management	0 1 2 3 4 5 6 7 8 9 10		Introduction To Computer Systems And Networks (Y3)
196	**performance tuning**	Scalable Software Design	0 1 2 3 4 5 6 7 8 9 10		Object Oriented Programming (Y3)
197	**disaster recovery planning**	Database Recovery Techniques	0 1 2 3 4 5 6 7 8 9 10		Advanced Database And Database Management Systems (Y4)
198	**system integration**	Software Application Maintenance	0 1 2 3 4 5 6 7 8 9 10		Object Oriented Programming (Y3)
199	**security monitoring**	Distributed System Security	0 1 2 3 4 5 6 7 8 9 10		Distributed Systems (Y4)
200	**database management**	Data Cleaning	0 1 2 3 4 5 6 7 8 9 10		Research Methodology (Y3)
201	**network administration**	Network Infrastructure Design	0 1 2 3 4 5 6 7 8 9 10		Networks Ii (Y4)
202	**ci/cd pipelines**	CI/CD Pipeline Implementation	0 1 2 3 4 5 6 7 8 9 10		Software Development And It Operations Devops (Y4)
203	**server management**	Security Solution Management	0 1 2 3 4 5 6 7 8 9 10		Information Security (Y5)
204	**network administration**	Network Devices	0 1 2 3 4 5 6 7 8 9 10		Network Security (Y5)
205	**security compliance**	Database Security Policy Design	0 1 2 3 4 5 6 7 8 9 10		Advanced Database And Database Management Systems (Y4)
206	**technical support**	Network Troubleshooting	0 1 2 3 4 5 6 7 8 9 10		Introduction To Computer Systems And Networks (Y3)
207	**vmware virtualization**	Virtualization With VMware Workstation	0 1 2 3 4 5 6 7 8 9 10		Systems And Networks Administration (Y4)

#	Market Skill	Course Learning Outcome	Score (0–10) — circle one	Note	Course (Year)
208	**system maintenance**	Distributed System Processes	0 1 2 3 4 5 6 7 8 9 10		Distributed Systems (Y4)
209	**it infrastructure management**	Network Infrastructure Design	0 1 2 3 4 5 6 7 8 9 10		Networks Ii (Y4)
210	**it infrastructure management**	Information Security Policy Management	0 1 2 3 4 5 6 7 8 9 10		Information Security (Y5)
211	**system maintenance**	Session Management	0 1 2 3 4 5 6 7 8 9 10		Internet Programming Ii (Y4)
212	**it infrastructure management**	Network Infrastructure Design	0 1 2 3 4 5 6 7 8 9 10		Introduction To Computer Systems And Networks (Y3)
213	**aws cloud platforms**	Virtualization With VMware Workstation	0 1 2 3 4 5 6 7 8 9 10		Systems And Networks Administration (Y4)
214	**restful api development**	Service-Oriented Architecture	0 1 2 3 4 5 6 7 8 9 10		Cloud Computing (Y5)
215	**server management**	Distributed Database Management	0 1 2 3 4 5 6 7 8 9 10		Advanced Database And Database Management Systems (Y4)
216	**server management**	Network Infrastructure Design	0 1 2 3 4 5 6 7 8 9 10		Networks Ii (Y4)
217	**system maintenance**	Network Maintenance and Troubleshooting	0 1 2 3 4 5 6 7 8 9 10		Network Design (Y4)
218	**mysql database**	Database Queries	0 1 2 3 4 5 6 7 8 9 10		Data Mining (Y5)
219	**security compliance**	C++ Programming	0 1 2 3 4 5 6 7 8 9 10		Algorithm And Programming (Y3)
220	**postgresql database**	Database Migrations	0 1 2 3 4 5 6 7 8 9 10		Internet Programming Ii (Y4)
221	**aws cloud platforms**	Cloud Computing Security	0 1 2 3 4 5 6 7 8 9 10		Network Security (Y5)
222	**security monitoring**	Cloud Computing Security	0 1 2 3 4 5 6 7 8 9 10		Cloud Computing (Y5)
223	**it infrastructure management**	Infrastructure as a Service (IaaS)	0 1 2 3 4 5 6 7 8 9 10		Cloud Computing (Y5)
224	**data analysis**	Distributed System Performance Analysis	0 1 2 3 4 5 6 7 8 9 10		Distributed Systems (Y4)
225	**server management**	Load Balancing	0 1 2 3 4 5 6 7 8 9 10		Cloud Computing (Y5)
226	**technical documentation**	Control Statements	0 1 2 3 4 5 6 7 8 9 10		Object Oriented Programming (Y3)
227	**network troubleshooting**	Network Troubleshooting	0 1 2 3 4 5 6 7 8 9 10		Introduction To Computer Systems And Networks (Y3)
228	**data analysis**	Object-Oriented Analysis and Design	0 1 2 3 4 5 6 7 8 9 10		Object Oriented Programming (Y3)
229	**disaster recovery planning**	Network Service Backup and Recovery Strategies	0 1 2 3 4 5 6 7 8 9 10		Systems And Networks Administration (Y4)
230	**firewall configuration**	VLAN Configuration	0 1 2 3 4 5 6 7 8 9 10		Networks Ii (Y4)
231	**network troubleshooting**	Security Solutions for System Support and Operations	0 1 2 3 4 5 6 7 8 9 10		Information Security (Y5)
232	**security monitoring**	Security Technology Deployment	0 1 2 3 4 5 6 7 8 9 10		Network Security (Y5)
233	**postgresql database**	Relational Database Design	0 1 2 3 4 5 6 7 8 9 10		Information System Analysis And Design (Y3)
234	**code reviews**	Huffman Coding	0 1 2 3 4 5 6 7 8 9 10		Image Processing (Y5)
235	**system maintenance**	Application Health Monitoring	0 1 2 3 4 5 6 7 8 9 10		Software Development And It Operations Devops (Y4)
236	**security compliance**	Laravel Security Features	0 1 2 3 4 5 6 7 8 9 10		Internet Programming Ii (Y4)
237	**disaster recovery planning**	System Reliability Engineering	0 1 2 3 4 5 6 7 8 9 10		Network Design (Y4)
238	**git version control**	Regular Expressions	0 1 2 3 4 5 6 7 8 9 10		Natural Language Processing (Y5)
239	**api integration**	Application Programming Interfaces (APIs)	0 1 2 3 4 5 6 7 8 9 10		Internet Programming Ii (Y4)
240	**project management**	Distributed System Processes	0 1 2 3 4 5 6 7 8 9 10		Distributed Systems (Y4)
241	**code reviews**	Scalable Software Design	0 1 2 3 4 5 6 7 8 9 10		Object Oriented Programming (Y3)
242	**technical support**	Security Solutions for System Support and Operations	0 1 2 3 4 5 6 7 8 9 10		Information Security (Y5)
243	**java programming**	Clustering	0 1 2 3 4 5 6 7 8 9 10		Introduction To Artificial Intelligence (Y5)
244	**docker containerization**	Containerization	0 1 2 3 4 5 6 7 8 9 10		Cloud Computing (Y5)
245	**aws cloud platforms**	Infrastructure Provisioning and Management with IaC	0 1 2 3 4 5 6 7 8 9 10		Software Development And It Operations Devops (Y4)

#	Market Skill	Course Learning Outcome	Score (0–10) — circle one	Note	Course (Year)
246	**system integration**	Integration Testing	0 1 2 3 4 5 6 7 8 9 10		Internet Programming Ii (Y4)
247	**performance tuning**	Algorithm Analysis	0 1 2 3 4 5 6 7 8 9 10		Discrete Mathematics (Y3)
248	**microsoft office proficiency**	C++ Programming	0 1 2 3 4 5 6 7 8 9 10		Algorithm And Programming (Y3)
249	**database management**	Database Migrations	0 1 2 3 4 5 6 7 8 9 10		Internet Programming Ii (Y4)
250	**performance tuning**	Image Smoothing	0 1 2 3 4 5 6 7 8 9 10		Image Processing (Y5)

Blind sheet: similarity scores, bands and job-demand counts are hidden to prevent priming (kept in the master file). Optional mapping: 0–3 = No · 4–6 = Partly · 7–10 = Yes. sheet 78dfd3fe9b63 · run 20260727T105055Z-475eb3

Glossary — market-skill terms (reference only)

If a skill is clearly unrelated to the course outcome, a low score (0–2) is the expected answer.

Term	Meaning
api integration	connecting software to other systems through their APIs (payment, mapping, messaging services)
aws cloud platforms	Amazon Web Services — cloud servers, storage and managed services
ci/cd pipelines	automated build-test-deploy chains (e.g. GitHub Actions, Jenkins) that ship code continuously
code reviews	systematically reading a colleague's code changes before they are merged
disaster recovery planning	preparing backup systems and procedures so IT services survive a major failure
docker containerization	packaging applications into portable Docker containers
firewall configuration	setting up network firewalls and their traffic rules
git version control	tracking and merging source-code changes with Git
incident response	the organised process of detecting, containing and recovering from a security incident
it infrastructure management	operating an organisation's servers, networks and core IT services
microsoft office proficiency	practical use of Word, Excel and PowerPoint
mysql database	MySQL — an open-source relational database system
performance tuning	measuring and improving the speed of systems, databases or code
postgresql database	PostgreSQL — an open-source relational database system
restful api development	building web APIs that follow the REST style (HTTP endpoints returning JSON)
security compliance	meeting security standards/policies (ISO 27001, PCI-DSS) and passing audits
security monitoring	watching logs/alerts (SIEM) to detect attacks in progress
system integration	making separate software systems work together as one
technical documentation	writing manuals, runbooks and design documents for technical systems
vmware virtualization	running many virtual machines on one physical server with VMware